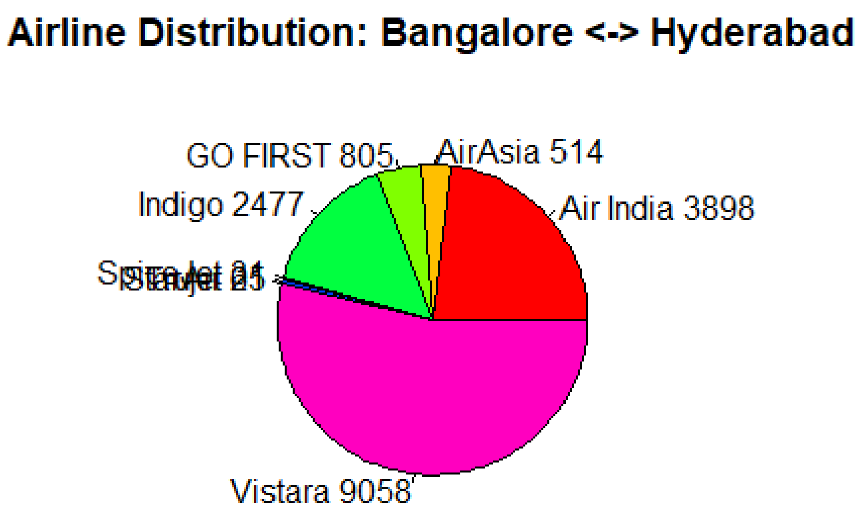
```r
library(readxl)
goibibo_flights_data <-
read_excel("C:/Users/mohni/Downloads/goibibo_flights_data.xlsx")

# Convert price to integer
goibibo_flights_data$price <- as.integer(gsub(",", "",
goibibo_flights_data$price))

# Filter for flights from Bangalore to Hyderabad
bangalore_to_hyderabad_prices <- goibibo_flights_data$price[
  goibibo_flights_data$from == "Bangalore" & goibibo_flights_data$to ==
"Hyderabad"
]

# Filter for flights from Hyderabad to Bangalore
hyderabad_to_bangalore_prices <- goibibo_flights_data$price[
  goibibo_flights_data$from == "Hyderabad" & goibibo_flights_data$to ==
"Bangalore"
]

# Calculate the median prices
median_price_bangalore_to_hyderabad <- median(bangalore_to_hyderabad_prices,
na.rm = TRUE)
median_price_hyderabad_to_bangalore <- median(hyderabad_to_bangalore_prices,
na.rm = TRUE)

# Print the median prices
print(paste("Median price of flights from Bangalore to Hyderabad:",
median_price_bangalore_to_hyderabad))

## [1] "Median price of flights from Bangalore to Hyderabad: 7891"

print(paste("Median price of flights from Hyderabad to Bangalore:",
median_price_hyderabad_to_bangalore))

## [1] "Median price of flights from Hyderabad to Bangalore: 6924"

# Extract airline data for Bangalore to Hyderabad
bangalore_to_hyderabad_airlines <- goibibo_flights_data$airline[
  goibibo_flights_data$from == "Bangalore" & goibibo_flights_data$to ==
"Hyderabad"
]

# Extract airline data for Hyderabad to Bangalore
hyderabad_to_bangalore_airlines <- goibibo_flights_data$airline[
  goibibo_flights_data$from == "Hyderabad" & goibibo_flights_data$to ==
"Bangalore"
]
```

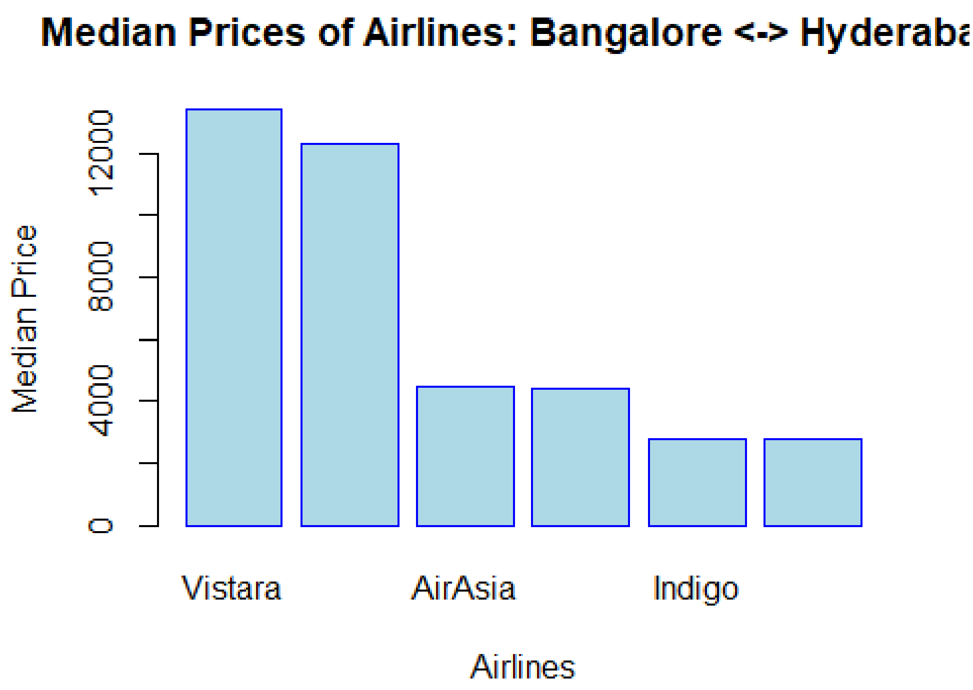
```r
# Combine the two sets of airlines
combined_airlines <- c(bangalore_to_hyderabad_airlines,
hyderabad_to_bangalore_airlines)

# Create a table of airline frequencies
airline_frequencies <- table(combined_airlines)

# Create a pie chart
pie(airline_frequencies,
    main = "Airline Distribution: Bangalore <-> Hyderabad",
    col = rainbow(length(airline_frequencies)),
    labels = paste(names(airline_frequencies), airline_frequencies))
```




```r
# Function to get median price for a specific airline
get_median_price <- function(airline_name) {
  # Filter prices for Bangalore to Hyderabad
  bangalore_to_hyderabad_prices <- goibibo_flights_data$price[
    goibibo_flights_data$from == "Bangalore" &
    goibibo_flights_data$to == "Hyderabad" &
    goibibo_flights_data$airline == airline_name
  ]

  # Filter prices for Hyderabad to Bangalore
  hyderabad_to_bangalore_prices <- goibibo_flights_data$price[
    goibibo_flights_data$from == "Hyderabad" &
    goibibo_flights_data$to == "Bangalore" &
```

```r
    goibibo_flights_data$airline == airline_name
  ]

  # Combine prices
  combined_prices <- c(bangalore_to_hyderabad_prices,
hyderabad_to_bangalore_prices)

  # Return median price
  return(median(combined_prices, na.rm = TRUE))
}

# List of airlines to check
airlines <- c("Vistara", "Air India", "AirAsia", "GO FIRST", "Indigo",
"SpiceJet")

# Create a data frame to store median prices
median_prices <- data.frame(Airline = airlines, MedianPrice = NA)

# Loop through each airline and get the median price
for (i in 1:length(airlines)) {
  median_prices$MedianPrice[i] <- get_median_price(airlines[i])
}

# View the median prices
print(median_prices)

##     Airline MedianPrice
## 1   Vistara       13391
## 2 Air India       12296
## 3   AirAsia        2765
## 4  GO FIRST        4497
## 5    Indigo        2764
## 6  SpiceJet        4381

airlines <- c("Vistara", "Air India", "AirAsia", "GO FIRST", "Indigo",
"SpiceJet")
median_prices <- c(13391, 12296, 4497, 4381, 2765,  2764)

# Create a data frame
median_prices_df <- data.frame(Airline = airlines, MedianPrice =
median_prices)

# Create the bar plot
barplot(
  median_prices_df$MedianPrice,
  names.arg = median_prices_df$Airline,
  main = "Median Prices of Airlines: Bangalore <-> Hyderabad",
  xlab = "Airlines",
  ylab = "Median Price",
  col = "lightblue",
```

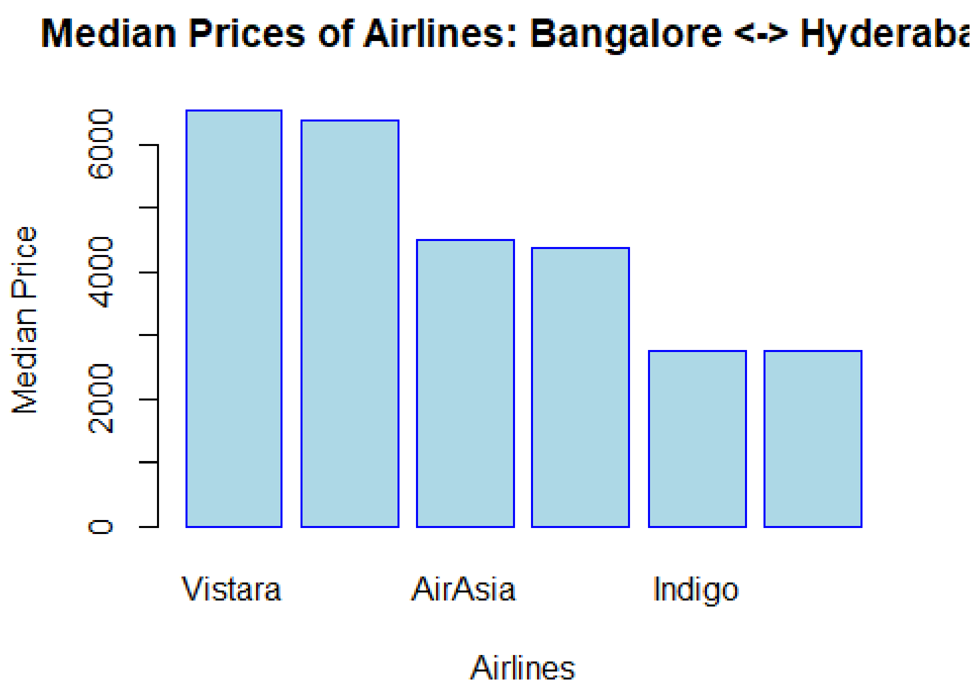
```
  border = "blue"
)
```




```
# Create a vector of target airlines
target_airlines <- c("Vistara", "Air India", "Indigo", "AirAsia", "GO FIRST",
"SpiceJet")


condition <- (goibibo_flights_data$from == "Bangalore" &
goibibo_flights_data$to == "Hyderabad") |
             (goibibo_flights_data$from == "Hyderabad" &
goibibo_flights_data$to == "Bangalore")

# Filter the dataset
filtered_flights <- goibibo_flights_data[goibibo_flights_data$airline %in%
target_airlines & condition, ]

# Create a contingency table for class counts
class_counts <- table(filtered_flights$airline, filtered_flights$class)

# Print the class counts
print(class_counts)

##
##               business economy
##   Air India      1509    2389
```

```
##   AirAsia          0     514
##   GO FIRST         0     805
##   Indigo           0    2477
##   SpiceJet         0      31
##   Vistara       4195    4863

# Convert the price column to integer
goibibo_flights_data$price <- as.integer(goibibo_flights_data$price)

# Extract prices for Vistara flights from Hyderabad to Bangalore
vistara_prices <- goibibo_flights_data$price[
  goibibo_flights_data$airline == "Vistara" &
  goibibo_flights_data$from == "Hyderabad" &
  goibibo_flights_data$to == "Bangalore"
]

# Extract prices for Air India flights from Hyderabad to Bangalore
airindia_prices <- goibibo_flights_data$price[
  goibibo_flights_data$airline == "Air India" &
  goibibo_flights_data$from == "Hyderabad" &
  goibibo_flights_data$to == "Bangalore"
]

# Calculate median prices for Vistara by class
vistara_business_prices <-
vistara_prices[goibibo_flights_data$class[goibibo_flights_data$airline ==
"Vistara" &

goibibo_flights_data$from == "Hyderabad" &

goibibo_flights_data$to == "Bangalore"] == "business"]

vistara_economy_prices <-
vistara_prices[goibibo_flights_data$class[goibibo_flights_data$airline ==
"Vistara" &

goibibo_flights_data$from == "Hyderabad" &

goibibo_flights_data$to == "Bangalore"] == "economy"]

# Calculate median prices for Air India by class
airindia_business_prices <-
airindia_prices[goibibo_flights_data$class[goibibo_flights_data$airline ==
"Air India" &

goibibo_flights_data$from == "Hyderabad" &

goibibo_flights_data$to == "Bangalore"] == "business"]
```

```r
airindia_economy_prices <-
airindia_prices[goibibo_flights_data$class[goibibo_flights_data$airline ==
"Air India" &

goibibo_flights_data$from == "Hyderabad" &

goibibo_flights_data$to == "Bangalore"] == "economy"]

# Finding and printing median values
vistara_business_median <- median(vistara_business_prices, na.rm = TRUE)
vistara_economy_median <- median(vistara_economy_prices, na.rm = TRUE)
airindia_business_median <- median(airindia_business_prices, na.rm = TRUE)
airindia_economy_median <- median(airindia_economy_prices, na.rm = TRUE)

# Print the results
print(paste("Vistara Business Median Price:", vistara_business_median))

## [1] "Vistara Business Median Price: 57553"

print(paste("Vistara Economy Median Price:", vistara_economy_median))

## [1] "Vistara Economy Median Price: 6526"

print(paste("Air India Business Median Price:", airindia_business_median))

## [1] "Air India Business Median Price: 57439"

print(paste("Air India Economy Median Price:", airindia_economy_median))

## [1] "Air India Economy Median Price: 6377"

airlines <- c("Vistara", "Air India", "AirAsia", "GO FIRST", "Indigo",
"SpiceJet")
median_prices <- c(6526, 6377, 4497, 4381, 2765,  2764)

# Create a data frame for better handling
median_prices_df <- data.frame(Airline = airlines, MedianPrice =
median_prices)

# Create the bar plot
barplot(
  median_prices_df$MedianPrice,
  names.arg = median_prices_df$Airline,
  main = "Median Prices of Airlines: Bangalore <-> Hyderabad",
  xlab = "Airlines",
  ylab = "Median Price",
  col = "lightblue",
  border = "blue"
)
```

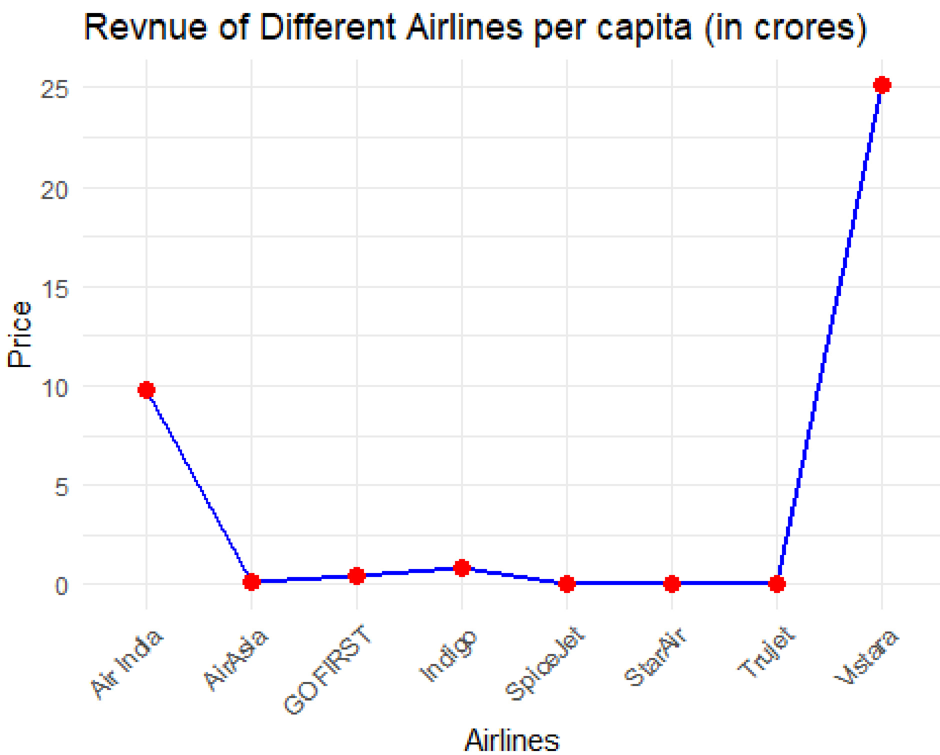



```r
# Filter for flights between Hyderabad and Bangalore
hyderabad_to_bangalore <- goibibo_flights_data[
  goibibo_flights_data$from == "Hyderabad" & goibibo_flights_data$to ==
"Bangalore", ]

bangalore_to_hyderabad <- goibibo_flights_data[
  goibibo_flights_data$from == "Bangalore" & goibibo_flights_data$to ==
"Hyderabad", ]

# Combine the two datasets
combined_flights <- rbind(hyderabad_to_bangalore, bangalore_to_hyderabad)

# Calculate revenue for each airline
revenue_per_airline <- aggregate(price ~ airline, data = combined_flights,
FUN = sum, na.rm = TRUE)

# Rename columns
colnames(revenue_per_airline) <- c("Airline", "TotalRevenue")

# Print the revenue summary
print(revenue_per_airline)

##     Airline TotalRevenue
## 1 Air India     97666231
## 2   AirAsia      1588293
## 3  GO FIRST      3893402
```

```
## 4    Indigo      8032657
## 5  SpiceJet       135811
## 6   StarAir       303903
## 7    Trujet        83580
## 8   Vistara    250864305
```

```
airline_data <- data.frame(
  Airline = c("Vistara","Air India", "AirAsia", "GO FIRST", "Indigo",
              "SpiceJet", "StarAir", "Trujet"),
  Price = c(25.09,9.77, 0.16, 0.39, 0.80,
           0.014, 0.030, 0.0084 )
)

library(ggplot2)

ggplot(data = airline_data, aes(x = Airline, y = Price, group = 1)) +
  geom_line(color = "blue", size = 1) +          # Line plot
  geom_point(color = "red", size = 3) +          # Add points
  theme_minimal() +                              # Clean theme
  labs(title = "Revnue of Different Airlines per capita (in crores)",  # Add
title
     x = "Airlines",                             # X-axis label
     y = "Price") +                              # Y-axis label
  theme(axis.text.x = element_text(angle = 45, hjust = 1))  # Rotate X-axis
labels
```

```
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## ℹ Please use `linewidth` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```




```
# Filter for flights originating from Bangalore
bangalore_destinations <- goibibo_flights_data[goibibo_flights_data$from ==
"Bangalore", ]

# Count the occurrences of each destination
destination_counts <- table(bangalore_destinations$to)

destination_counts

##
##    Chennai      Delhi  Hyderabad    Kolkata     Mumbai
##       6410      13756       8971      10029      12940

pie(destination_counts,
    main = "Flight Destinations from Bangalore",
    col = rainbow(length(destination_counts)),
    labels = paste(names(destination_counts), destination_counts))
```

Flight Destinations from Bangalore